Top-two-layer spectra: blue = species present (mean ± 1 SD); dashed = species absent mean

CHO: n=5
CO: n=30
CO2: n=3 (sparse)
COCHO: n=1 (sparse)
COH: n=6
COOH: n=8
H: n=22
O: n=7
OCCOH: n=4 (sparse)
OCHO: n=8
OH: n=5

Unit-area absorption (1/eV)

0.000
0.005
0.010
0.015
0.020

8980
8990
9000
9010
9020
9030

Photon energy (eV)

Shading describes variation across configurations, not adsorption energy. Mixed structures contribute to each species present.